ON SOME PROBLEMS OF ORTHOMORPHOSIS (continued)

[921, pp. 204–205]

giving

$$z_1 = \frac{A(z-a)}{1-az},$$

which, for $z\bar{z}=1$, becomes

$$z_1\bar{z}_1 = A\bar{A}\,\frac{z-a}{1-az}\cdot\frac{\bar{z}-a}{1-a\bar{z}},$$

so that ($A\bar{A}$ being $=1$) this gives $z_1\bar{z}_1 = 1$.

16. This is a solution with three arbitrary constants, viz. A (which may be put $=\cos\lambda + i\sin\lambda$) is a single arbitrary constant, and $\alpha, = a+ib$, is two arbitrary constants; and these constants may be so determined that a given point in the interior of the one circle, and a given point on the circumference thereof, shall correspond respectively to a given point in the interior of the other circle and to a given point on the circumference thereof. According to a well-known theorem of Riemann's, any two simply connected areas included within given closed curves respectively may be made to correspond to each other, and that in one way only, under the foregoing conditions as to a pair of interior points and a pair of boundary points; and we have, in what just precedes, the solution of the problem in the case of two equal circles.

17. In the case of any other solution, we thus know that the correspondence between the two circumferences cannot and does not imply a (1, 1) correspondence between the areas of the two circles: but it is interesting to enquire what happens. I take a very particular case

$$z_1 = \frac{z(z-2)}{1-2z},$$

and therefore

$$\bar{z}_1 = \frac{\bar{z}(\bar{z}-2)}{1-2\bar{z}},$$

and consequently

$$z_1\bar{z}_1 = \frac{z\bar{z}\,\{z\bar{z}-2(z+\bar{z})+4\}}{1-2(z+\bar{z})+4z\bar{z}},$$

that is,

$$x_1^2+y_1^2 = \frac{(x^2+y^2)\,\{x^2+y^2-4x+4\}}{4(x^2+y^2)-4x+1},$$

so that, writing $x_1^2+y_1^2-1=0$, we have $x^2+y^2-1=0$, a correspondence of the two circumferences. But to the circumference $x_1^2+y_1^2-1=0$ there corresponds not only the circumference $x^2+y^2-1=0$, but another circumference. In fact, writing $x_1^2+y_1^2=1$, we have

$$4(x^2+y^2)-4x+1 = (x^2+y^2)(x^2+y^2-4x+4),$$

that is,

$$(x^2+y^2)^2 - 4x(x^2+y^2) + 4x - 1 = 0,$$

or

$$(x^2+y^2-1)(x^2+y^2-4x+1) = 0,$$

and there is thus the other circle

$$x^2+y^2-4x+1=0, \quad \text{or say } (x-2)^2+y^2-3=0,$$

viz. this is a circle, coordinates of centre $(2,0)$ and radius $=\sqrt{3}$, cutting the circle $x^2+y^2-1=0$ in two real points. Referring to the figures 6 (xy) and 7 (x_1y_1), and observing that $x_1=0$,

$y_1 = 0$, that is, $z_1 = 0$, gives $z = 0$, or $z = 2$, that is, the points $x = 0$, $y = 0$, and $x = 2$, $y = 0$, we see that to the centre O in figure 7, there correspond in figure 6 the points O, M which are the centres of the two circles. To any small closed curve, or say any small circle surrounding the point O of figure 7, there correspond in figure 6 small closed curves surrounding the points O, M respectively; and if in figure 7 the radius of the circle continually increases and becomes nearly equal to unity, the closed curves of figure 6 continually increase, changing at the same time their forms, and assume the forms shown by the dotted lines of figure 6. It thus appears that, to the whole area of the circle $x_1^2 + y_1^2 - 1 = 0$ of figure 7, there correspond the two lunes ACB and ABD of figure 6; or if we attend only to the area included within the circle $x^2 + y^2 - 1 = 0$ of this figure, then there corresponds not the whole area of this circle, but only the area of the lune ACB: and thus that the assumed relation $z_1 = \dfrac{z(z-2)}{1-2z}$ establishes, in fact, an orthomorphosis of the circle $x_1^2 + y_1^2 - 1 = 0$ into the lune ACB which lies inside the circle $x^2 + y^2 - 1 = 0$ and outside the circle $(x-2)^2 + y^2 - 3 = 0$. It may be added that, to the infinite area outside the circle $x_1^2 + y_1^2 - 1 = 0$ of figure 7, there correspond in figure 6 first the area of the lens AB common to the two circles, and secondly the area outside the two circles: we have thus an orthomorphosis of the area outside the circle $x_1^2 + y_1^2 - 1 = 0$ into these two areas respectively.

A somewhat more elegant example would have been that of the correspondence

$$z_1 = \frac{z(z-\sqrt{2})}{1-\sqrt{2}\,z};$$

here, corresponding to the circumference $x_1^2 + y_1^2 - 1 = 0$, we have the two equal circumferences $x^2 + y^2 - 1 = 0$, and $(x-\sqrt{2})^2 + y^2 - 1 = 0$: and to the whole area of the circle $x_1^2 + y_1^2 - 1 = 0$, there correspond two equal lunes ACB and ABD.

NOTE ON THE LUNAR THEORY.

[922]

From the *Monthly Notices of the Royal Astronomical Society*, vol. LII. (1892), pp. 2–5.

In the Lunar Theory, in whatever way worked out, the values ultimately obtained for the coordinates r, v, y should of course satisfy identically the equations of motion; and that they do so, is the ultimate verification of the correctness of the results obtained. It can hardly be hoped for that such a verification will ever be made for Delaunay's results; and yet it would seem generally that the labour of such a verification of the results to any extent, while exceeding (and possibly greatly exceeding) that of obtaining these results by any method employed for that purpose, ought still to be, so to speak, a labour of the same order. And one can, moreover, imagine the process of verification so arranged as to be a process of mere routine which could be carried out by ordinary computers. But, however this may be, I think it is not without interest to exhibit the verification to a very small extent, viz. to e, m^4.

I think there is an advantage in using capital letters for the arguments, and I accordingly write G (instead of Delaunay's g), to denote the mean anomaly.

The equations of motion are:

$$\frac{d}{dt}\left(\frac{dr}{dt}\right) - r\left\{\cos^2 y\left(\frac{dv}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2\right\} + \frac{m^2 r}{r^2} = \frac{d\Omega}{dr},$$

$$\frac{d}{dt}\left(r^2\cos^2 y\,\frac{dv}{dt}\right) = \frac{d\Omega}{dv},$$

$$\frac{d}{dt}\left(r^2\frac{dy}{dt}\right) + r^2\sin y\cos y\left(\frac{dv}{dt}\right)^2 = \frac{d\Omega}{dy},$$

$$\Omega = \frac{m'r^2}{r'^3}\left\{\tfrac{3}{2}\cos^2 y\cos^2(v-v') - \tfrac{1}{2}\right\};$$

or say

$$\Omega = \frac{m'r^2}{r'^3}\left\{\tfrac{3}{2}\cos^2 y\cos^2(v-v') - \tfrac{1}{2}\right\};$$

and thus the equations become

$$\frac{d^2r}{dt^2} - r\left\{\cos^2 y\left(\frac{dv}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2\right\} + \frac{m^2}{r^2} = \frac{m'^2 r}{r'^3}\left\{3\cos^2 y\cos^2(v-v') - 1\right\},$$

$$\frac{d}{dt}\left(r^2\cos^2 y\frac{dv}{dt}\right) = -\frac{3m'^2r^2}{r'^3}\cos^2 y\sin(v-v')\cos(v-v'),$$

$$\frac{d}{dt}\left(r^2\frac{dy}{dt}\right) + r^2\sin y\cos y\left(\frac{dv}{dt}\right)^2 = -\frac{3m'^2r^2}{r'^3}\sin y\cos y\cos^2(v-v').$$

To simplify as much as possible, take the Sun's orbit to be circular, i.e. $r' = a'$, $v' = mnt$; also neglect y^2: the first and second equations are

$$\frac{d^2r}{dt^2} - r\left(\frac{dv}{dt}\right)^2 + \frac{m^2}{r^2} = m'^2 r\left\{\tfrac{1}{2} + \tfrac{3}{2}\cos(2v-2v')\right\},$$

$$\frac{d}{dt}\left(r^2\frac{dv}{dt}\right) = -\tfrac{3}{2}m'^2r^2\sin(2v-2v'),$$

and if for convenience of working we write $a = 1$, $n = 1$, then the first equation may be written

$$\frac{d^2r}{dt^2} - r\left(\frac{dv}{dt}\right)^2 + \frac{1}{r^2} = m^2 r\bigl\{\tfrac{1}{2} + \tfrac{3}{2}\cos(2v - 2v')\bigr\}\,,$$

and similarly the second equation is

$$\frac{d}{dt}\left(r^2\frac{dv}{dt}\right) = -\tfrac{3}{2}m^2r^2\sin(2v - 2v'),$$

and the third equation may be disregarded.

The two equations should be satisfied by Delaunay's values, putting therein $e' = 0$, $y = 0$; say by the values

$$\begin{aligned}\frac{1}{r} &= 1 + \tfrac{1}{2}m^2 + \tfrac{1}{16}m^4 \\ &+ \bigl(m^2 + \tfrac{11}{8}m^3 + \tfrac{59}{18}m^4\bigr)\cos 2D \\ &+ e\cos G \\ &+ e\bigl(\tfrac{15}{8}m - \tfrac{19}{16}m^2\bigr)\cos(2D + G) \\ &+ e\bigl(\tfrac{1}{2}m + \tfrac{17}{16}m^2\bigr)\cos(2D - G)\end{aligned}$$

$$r\frac{dv}{dt} = 1 + 2e\cos G + e\bigl(\tfrac{15}{8}m - \cdots\bigr)\sin G\ +\ \cdots$$

where

$$D = (1 - m)nt, \qquad G = cnt.$$

The verification for the first equation is

$$-\frac{1}{r^2}\frac{d^2r}{dt^2}\ -\ \left(\frac{dv}{dt}\right)^2\ +\ \frac{1}{r^2}\cdot\frac{1}{r}\ -\ m^2r\bigl\{\tfrac{1}{2} + \tfrac{3}{2}\cos(2v - 2v')\bigr\} = 0$$

	A	B	C	D	total
Const.	-1	$+1$			$= 0$
	$+\frac{1}{2}m^2$	$-\frac{1}{2}m^2$			$= 0$
	$+\frac{1}{16}m^4$	$-\frac{1}{16}m^4$			$= 0$
	$+2m^4$	$-\frac{1}{4}m^4$	$-\frac{15}{16}m^4$	$-\frac{3}{8}m^4$	$= 0$
$\cos 2D$	$+4m^2$	$-\frac{9}{2}m^2$	$+3m^2$	$-\frac{3}{2}m^2$	$= 0$
$\cos 4D$				$-\frac{15}{16}m^4$	$= 0$
$\cos G$	e	$-\frac{1}{2}e$	$\frac{3}{2}e$		$= 0$
	$-\frac{3}{4}em^2$	$+3em^2$	$-\frac{3}{2}em^2$		$= 0$
$\cos(2D - G)$	$em(\frac{1}{2})$	$em(-\frac{1}{4})$	$em(-\frac{1}{4})$		$= 0$

The verification for the second equation is

$$\frac{2}{r}\frac{dr}{dt}\,r\frac{dv}{dt} + r^2\frac{d^2v}{dt^2} + \tfrac{3}{2}m^2\sin(2v - 2v') = 0$$

$\sin 2D$	$+4m^2$	$-\frac{11}{2}m^2$	$+\frac{3}{2}m^2$	$= 0$
$\sin 4D$	$+\frac{15}{8}m^4$	$-\frac{59}{16}m^4$	$+\frac{53}{32}m^4$	$= 0$
$\sin G$	$e\cdot 2$	$+e\cdot(-2)$		$= 0$
	$e\cdot 3m^2$	$+e\cdot 3m^2$		$= 0$
$\sin(2D + G)$	$e\cdot\frac{3}{4}m^2$	$e\cdot(-\frac{19}{4})m^2$	$e\cdot 3m^2$	$= 0$
$\sin(2D - G)$	$e\cdot\frac{1}{2}m$	$e\cdot(-\frac{1}{2})m$		$= 0$

and the verification is thus completed.

The following intermediate results may be recorded; $\dfrac{1}{r} = 1 + x$,

$$y^2 = \cdots\,, \quad -\log r = \cdots\,, \quad \frac{dv}{dt} = \cdots\,,$$

$\cos 2D$	$\frac{1}{2}m^2$	$\frac{1}{2}m^2 - \frac{15}{16}m^4$
$\cos 4D$	$+\frac{1}{8}m^4$	$\frac{1}{4}m^4$
$\cos G$	$+\frac{1}{2}em^2$	$e(1 - \frac{1}{2}m^2)$
$\cos(2D - G)$		
$\sin 2D$	$-2m^2 + \cdots$	
$\sin 4D$		
$\sin G$		
$\sin(2D - G)$	e	

These were, in fact, made use of for finding the foregoing values of r, $\dfrac{dv}{dt}$, &c.

NOTE ON A HYPERDETERMINANT IDENTITY.

[923]

From the *Messenger of Mathematics*, vol. XXI. (1892), pp. 131, 132.

The following is in effect a well-known theorem; but I am not sure whether it has been stated in a form at once so general and so precise.

If

$$\Omega = \phi(x_1, y_1)^A (x_2, y_2)^B (x_3, y_3)^C (x_4, y_4)^D \cdots$$

be a function separately homogeneous, and of the degrees A, B, C, D, ... in the sets of variables (x_1, y_1), (x_2, y_2), (x_3, y_3), (x_4, y_4), ... respectively; and if

$$12 = \xi_1\eta_2 - \xi_2\eta_1 = \partial_{x_1}\partial_{y_2} - \partial_{x_2}\partial_{y_1}, \quad \&c.,$$

then

$$(A\cdot 23 + B\cdot 31 + C\cdot 12)\Omega = 0,$$

when the variables (x_1, y_1), (x_2, y_2), (x_3, y_3), $(x_4, y_4), \ldots$, or only the variables (x_1, y_1), (x_2, y_2), (x_3, y_3) are therein severally replaced by (x, y).

In fact, we have

$$A\Omega = (x_1\xi_1 + y_1\eta_1)\Omega, \quad B\Omega = (x_2\xi_2 + y_2\eta_2)\Omega, \quad C\Omega = (x_3\xi_3 + y_3\eta_3)\Omega;$$

thus the expression is

$$\{(x_1\xi_1 + y_1\eta_1)\cdot 23 + (x_2\xi_2 + y_2\eta_2)\cdot 31 + (x_3\xi_3 + y_3\eta_3)\cdot 12\}\Omega,$$

and if we herein replace the variables (x_1, y_1), (x_2, y_2), (x_3, y_3) in so far as they appear explicitly by (x, y), the expression becomes

$$= \{(x\xi_1 + y\eta_1)\cdot 23 + (x\xi_2 + y\eta_2)\cdot 31 + (x\xi_3 + y\eta_3)\cdot 12\}\Omega,$$

where the factor in { }, substituting for 23, 31, 12 their values $\xi_2\eta_3 - \xi_3\eta_2$, $\xi_3\eta_1 - \xi_1\eta_3$, $\xi_1\eta_2 - \xi_2\eta_1$, becomes identically $= 0$. The value of the expression is thus $= 0$; and of

course it remains $= 0$ when, consequently upon the foregoing change (x_1, y_1), (x_2, y_2), (x_3, y_3) each into (x, y), we also change (ξ_1, η_1), (ξ_2, η_2), (ξ_3, η_3) each into (ξ, η); and if we also change (x_4, y_4), &c., into (x, y).

It is clear that Ω may denote the covariant symbol

$$\Omega = 23^\alpha\, 31^\beta\, 12^\gamma\, 14^\delta\, 24^\varepsilon\, 34^\zeta \cdots U_1 V_2 W_3 T_4 \cdots,$$

where U, V, W, T, ... denote quantics

$$(a, \ldots |x, y)^m, \quad (a', \ldots |x, y)^n, \quad \ldots$$

of the degrees m, n, p, q, ... respectively, and U_1, V_2, &c., are the corresponding functions $(a, \ldots |x_1, y_1)^m$, $(a', \ldots |x_2, y_2)^n$, &c., the values of A, B, C being here

$$A = m - \beta - \gamma - \delta - \cdots,$$

$$B = n - \gamma - \alpha - \varepsilon - \cdots,$$

$$C = p - \alpha - \beta - \zeta - \cdots;$$

the theorem expresses the property that the covariants $12\,\Omega$, $23\,\Omega$, $31\,\Omega$, are linearly connected together; or, writing it in the form $(A\cdot 12 - C\cdot 13 + B\cdot 23)\Omega = 0$, we have the proper linear combination $A\cdot 12\,\Omega - C\cdot 13\,\Omega$ of the two covariants $12\,\Omega$ and $13\,\Omega$, equal to $-B\cdot 23\,\Omega$, a determinate multiple of $23\,\Omega$. Speaking roughly, we say that the difference of the covariants $12\,\Omega$ and $13\,\Omega$ is equal to $23\,\Omega$.

ON THE NON-EXISTENCE OF A SPECIAL GROUP OF POINTS.

[924]

From the *Messenger of Mathematics*, vol. XXI. (1892), pp. 132, 133.

It is well known that, taking in a plane any eight points, every cubic through these passes through a determinate ninth point. It is interesting to show that there is no system of seven points such that every cubic through these passes through a determinate eighth point.

Assuming such a system: first, no three of the points can be in a line: for, if they were, then among the cubics through the seven points we have the line through the three points and an arbitrary conic through the remaining four points, and these composite cubics have no common eighth point of intersection.

Secondly, no six of the points can be on a conic: for, if they were, then among the cubics through the seven points we have the conic through the six points and an arbitrary line through the remaining point, and these composite cubics have no common eighth point of intersection.

Taking now the points to be 1, 2, 3, 4, 5, 6, 7; among the cubics through these, we have the composite cubics (A, P), (B, Q), (C, R), where A, B, C are the lines 67, 75, 56, and P, Q, R the conics 12345, 12346, 12347 respectively; by what precedes, the points 5, 6, 7 do not lie on a line, and the points (6, 7), (7, 5) and (5, 6) neither of them lie on the conics P, Q, R respectively.

The common eighth point of intersection, if it exists, must be

$$(A, B, C); \quad (A, Q, R),\ (B, R, P),\ (C, P, Q);$$

$$(P, B, C),\ (Q, C, A),\ (R, A, B);\ \text{or}\ (P, Q, R).$$

There is no point (A, B, C).

There is no point (A, Q, R): for Q, R intersect only in the points 1, 2, 3, 4, no one of which lies on A; and similarly, there is no point (B, R, P) or (C, P, Q).

B, C intersect in 5, which is a point on P; and thus 5 is the only point (P, B, C). Similarly, 6 is the only point (Q, C, A) and 7 is the only point (R, A, B).

P, Q intersect in 1, 2, 3, 4, which are each of them on R; hence 1, 2, 3, 4 are the only points (P, Q, R).

Hence the points 1, 2, 3, 4, 5, 6, 7 present themselves each once, and only once, among the intersections of the three cubics, and there is no common eighth point of intersection.

ON WARING'S FORMULA FOR THE SUM OF THE mTH POWERS OF THE ROOTS OF AN EQUATION.

[925]

From the *Messenger of Mathematics*, vol. XXI. (1892), pp. 133–137.

The formula in question, Prob. I. of Waring's *Meditationes Algebraicae*, Cambridge, 1782, making therein a slight change of notation, is as follows: viz. the equation being

$$x^m + bx^{m-1} + cx^{m-2} + dx^{m-3} + \cdots = 0,$$

then we have

$$(-)^m S_{\alpha^m} =$$

$$\begin{aligned}
&b^m + \cdot 1\\
&\quad - mc \cdot b^{m-2} \;+\; \tfrac{1}{2} m \,.\, m-3 \cdot c^2 \cdot b^{m-4}\\
&\quad - md \cdot b^{m-3} \;+\; m \,.\, m-4 \cdot cd \cdot b^{m-5}\\
&\quad - me \cdot b^{m-4} \;+\; \tfrac{1}{2} m \,.\, m-5 \cdot (d^2 + 2ce) \cdot b^{m-6}\\
&\quad - mf \cdot b^{m-5} \;+\; m \,.\, m-6 \cdot (de + cf) \cdot b^{m-6}\\
&\quad - mg \cdot b^{m-6} \;+\; m \,.\, m-7 \cdot df \cdot b^{m-7}\\
&\quad \cdots\\
&\quad - \tfrac{1}{6} m \,.\, m-4 \,.\, m-5 \cdot c^3 \cdot b^{m-6}\\
&\quad + \tfrac{1}{6} m \,.\, m-5 \,.\, m-6 \cdot c^2 d \cdot b^{m-7}\\
&\quad - \tfrac{1}{6} m \,.\, m-6 \,.\, m-7 \cdot c^2 e \cdot b^{m-8}\\
&\quad - \tfrac{1}{6} m \,.\, m-6 \,.\, m-7 \cdot cd^2 \cdot b^{m-8}\\
&\quad + \tfrac{1}{24} m \,.\, m-5 \,.\, m-6 \,.\, m-7 \cdot c^4 \cdot b^{m-8}\\
&\quad + \&c.,
\end{aligned}$$

where, reckoning the weights of b, c, d, e, $\ldots$ as 1, 2, 3, 4, $\ldots$, respectively, the several terms are all the terms of the weight m, or (what is the same thing) in the coefficient of $b^{m-\theta}$ we have all the combinations of c, d, e, $\ldots$, (or say all the non-unitary combinations) of the weight θ, and where the numerical coefficient of

$$b^{m-\theta} c^\gamma d^\delta e^\varepsilon \cdots \quad (\gamma + \delta + \varepsilon + \cdots = \theta),$$

is

$$(-)^{\gamma+\delta+\varepsilon+\cdots} \frac{m \,.\, m - (\theta - \Sigma + 1) \,.\, m - (\theta - \Sigma + 2) \cdots m - (\theta - 1)}{\Pi\gamma \,.\, \Pi\delta \,.\, \Pi\varepsilon \cdots}$$

(Π being the factorial). Thus for the term $b^{m-8}c^2e^1$, $\theta = 8$; γ, δ, $\varepsilon = 2, 4, 1$ respectively (the other exponents each vanishing), and the coefficient is

$$(-)^3 \frac{m \,.\, m-6 \,.\, m-7}{1 \cdot 2 \cdot 1} = -\tfrac{1}{2} m \,.\, m-6 \,.\, m-7,$$

as above; and so in other cases.

For the MacMahon form

$$1 + bx + \tfrac{c}{1\cdot 2} x^2 + \tfrac{d}{1\cdot 2\cdot 3} x^3 + \cdots = (1 - ax)(1 - \beta x)(1 - \gamma x) \cdots,$$

or say

$$1 + bx + \tfrac{c}{1\cdot 2} x^2 + \cdots = (1 - ax)(1 - \beta x) \cdots$$

we must for b, c, d, ..., write b, $\dfrac{c}{1\cdot 2}$, $\dfrac{d}{1\cdot 2\cdot 3}$, $\dfrac{e}{1\cdot 2\cdot 3\cdot 4}$, ... respectively: we thus have

$$\begin{aligned}(-)^m S_{\alpha^m} = b^m & \\ & + b^{m-1}\cdot(-mc) \\ & + b^{m-2}\cdot\Big(-\tfrac{m}{1\cdot 2}d + \tfrac{1}{2}m\,.\,m-3\cdot\big(\tfrac{c}{1\cdot 2}\big)^2\Big) \\ & + b^{m-3}\cdot\big(-\tfrac{m}{1\cdot 2\cdot 3}e + m\,.\,m-4\cdot\tfrac{c}{1\cdot 2}\cdot\tfrac{d}{1\cdot 2\cdot 3}\big) \\ & + \&c.,\end{aligned}$$

or say

$$\begin{aligned}(-)^m\Pi(m-1)S_{\alpha^m} = \Pi(m-1)\Big[b^m - \Pi m\,\frac{c}{1\cdot 2}\,b^{m-2} + \cdots\Big] & \\ +\Pi m\,\frac{d}{1\cdot 2\cdot 3}\,b^{m-3} & \\ +\Pi m\,\frac{e}{1\cdot 2\cdot 3\cdot 4}\,b^{m-4} & \\ +\&c., &\end{aligned}$$

the numerical coefficient of

$$b^{m-\theta}\,c^\gamma\,d^\delta\,e^\varepsilon\cdots\qquad(\gamma+\delta+\varepsilon+\cdots=\theta)$$

being

$$(-)^{\gamma+\delta+\varepsilon+\cdots}\,\frac{\Pi m\,.\,m-(\theta-\Sigma+1)\,.\,m-(\theta-\Sigma+2)\cdots m-(\theta-1)}{\Pi\gamma\,.\,\Pi\delta\,.\,\Pi\varepsilon\cdots\,(\Pi 2)^\gamma(\Pi 3)^\delta(\Pi 4)^\varepsilon\cdots}.$$

It is convenient to write down the literal terms in alphabetical order (AO), calculating and affixing to each term the proper numerical coefficient; thus taking

$$1+bx+\frac{c}{1\cdot 2}x^2+\frac{d}{1\cdot 2\cdot 3}x^3+\cdots=(1-ax)(1-\beta x)(1-\gamma x)\cdots,$$

we find

$$\begin{aligned}-120S_6 = {} & g\ \cdot\ 1 \\ & bf\ \cdot\ -6 \\ & ce\ \cdot\ -15 \\ & d^2\ \cdot\ -10 \\ & b^2e\ \cdot\ +30 \\ & bcd\ \cdot\ +120 \\ & c^3\ \cdot\ +30 \\ & b^3d\ \cdot\ -120 \\ & b^2c^2\ \cdot\ -270 \\ & b^4c\ \cdot\ +360 \\ & b^6\ \cdot\ -120 \\ & \pm\ 541\end{aligned}$$

this expression, as representing the value of the non-unitary function S_6, being in fact a seminvariant.

It is to be remarked that the foregoing expression for the sum of the mth powers of the roots of the equation

$$x^m+bx^{m-1}+cx^{m-2}+\cdots=0$$

is, in fact, the series for x^m continued so far only as the exponent of b is not negative: see as to this Note XI. of Lagrange's *Equations Numeriques*. For the *a posteriori* verification, observe that we have

$$x = -b + \frac{c}{x} - \frac{d}{x^2} + \frac{e}{x^3} - \cdots,$$

or writing for a moment $u = -b$, say this

$$x = u + fx,$$

where

$$fx = \frac{c}{x} - \frac{d}{x^2} + \frac{e}{x^3} - \&c.$$

Hence, by Lagrange's theorem,

$$\begin{aligned} x^m = u^m + \frac{m\,u^m}{u}\left(\tfrac{c}{u} + \tfrac{-d}{u^2} + \tfrac{e}{u^3} + \cdots\right) \\ + \tfrac{1}{1\cdot 2}\left(\tfrac{c}{u} + \tfrac{-d}{u^2} + \tfrac{e}{u^3} + \cdots\right)' \cdot \frac{m\,u^{m-1}}{1\cdot 2} \\ + \tfrac{1}{1\cdot 2\cdot 3}(\cdots)'' \cdot \frac{m\,u^{m-2}}{1\cdot 2\cdot 3} \\ + \&c., \end{aligned}$$

where the accents denote differentiations in regard to u. This is

$$\begin{gathered} u^m + m\{c\,u^{m-2} + d\,u^{m-3} + e\,u^{m-4} + f\,u^{m-5} + \cdots\} \\ + \tfrac{1}{2}m\{(m-3)c^2u^{m-4} + (m-4)\cdot 2cd\cdot u^{m-5} + (m-5)(d^2+2ce)u^{m-6} + \cdots\} \\ - \tfrac{1}{6}m\{(m-4)(m-5)c^3u^{m-6} + \cdots\} \\ + \&c. \\ = u^m \\ + u^{m-1}\cdot(-mc) \\ + u^{m-2}\cdot(-md) \\ + u^{m-3}\cdot(-me) + \tfrac{1}{2}m\,.\,m-3\,.\,c^2 \\ + u^{m-4}\cdot(-mf) + \tfrac{1}{2}m\,.\,m-4\,.\,2cd \\ + u^{m-5}\cdot(-mg) + \tfrac{1}{2}m\,.\,m-5\,.\,(d^2+2ce) - \tfrac{1}{6}m\,.\,m-4\,.\,m-5\,.\,c^3 \\ + \&c., \end{gathered}$$

which, putting therein $u = -b$ and multiplying each side by $(-)^m$, is the before-mentioned formula for $(-)^m S_{\alpha^m}$: in that formula the series being continued only so far as the exponent of b is not negative.

I notice also that we cannot easily, by means of the known formula

$$S_{\alpha^m\beta^p} = S_{\alpha^m}S_{\alpha^p} - S_{\alpha^{m+p}},$$

deduce an expression for $S_{\alpha^m\beta^p}$: in fact, forming the product of the series for S_{α^m}, S_{α^p} respectively, this product is identically equal to the series for $S_{\alpha^{m+p}}$, or we seem to obtain $0 = S_{\alpha^m}S_{\alpha^p} - S_{\alpha^{m+p}}$: to obtain the correct formula, we have to take each of the three series only so far as the exponent of b therein respectively is not negative: and it is not easy to see how the resulting formula is to be expressed.

CORRECTED SEMINVARIANT TABLES FOR THE WEIGHTS 11 AND 12.

[926]

From the *American Journal of Mathematics*, vol. XIV. (1892), pp. 195–200.

The tables in my paper, "Seminvariant Tables." *American Journal of Mathematics*, vol. VII. (1885), pp. 59–73, [831], are not in the best form, but the deviations present themselves only in a few columns of the tables for the weights 11 and 12, viz. in the former of these two columns, and in the latter a single column, ought to be replaced by linear combinations of other columns; there are, besides, columns which should be new named; in regard hereto, there is a point of theory which requires to be made clear. I remark that in each table the literal terms are in alphabetical order (AO); this is the proper order for the final terms, and although (as about to be explained) the proper order for the initial terms is the counter order (CO), yet as the tables cannot be at the same time arranged in the one and the other order, I adhere to the AO as the proper arrangement for the terms of the tables; we have, however, to introduce the notion that, in general, it is not the top term of a column which is to be regarded as the initial term, and in connexion herewith to consider how the columns are to be named. An instance first presents itself for the weight 11: we have, see column 5 of the table for that weight here given, a seminvariant

$$\begin{array}{ll} fg & +1 \\ bf^2 & \\ bcg & \\ bdh & \\ beg & -5 \\ bf^2 & -3 \\ c^2h & -16 \\ & \\ b^3e^2 & +70; \end{array}$$

this is to be regarded, not as a seminvariant $fg - b^3e^2$ (it is hardly necessary to remark that, here and elsewhere the $-$ is not a minus sign, but a mere stroke), but as a seminvariant $c^2h - b^3e^2$, viz. c^2h is a term entering into the seminvariant, and which, although it is in AO subsequent, it is in CO precedent, to the terms fg, bcg and bf^2. The seminvariant contains the term $-16\,c^2h$ and other terms with the letter h, and it is a misnomer to call it $fg - b^3e^2$, a name implying that the highest letter thereof is g. Instead of the stroke, it would perhaps be better to write ∞, for instance $c^2h\infty b^3e^2$, where of course ∞ would be used as a mere conventional symbol.

For greater clearness, I give here the express definition of counter order, (CO), viz. whereas in AO we begin with the lowest letters, in CO we begin contrariwise with the highest letters. A term containing a higher letter or higher power of such letter precedes a term containing a lower letter or lower power of the same letter — or in the easiest form, the counter order is the alphabetical order corresponding to the reversed arrangement $z, y, \ldots, f, e, d, c, b$ of the letters.

A symbol, as $c^2h - b^3e^2$ above, may be regarded as referring to a set of terms c^2h, b^3e^2 and all the terms which are in CO subsequent to c^2h and in AO precedent to b^3e^2: as by supposition the terms are arranged in AO, the set includes no term lower than b^3e^2, or say the bottom term b^3e^2 is also the final term of the set, but it does include terms fg, bcg and bf^2 higher than c^2h, and thus the top term fg is not, but c^2h is, the initial term of the set. It should be remarked that a seminvariant $c^2h - b^3e^2$ need not include all the terms of the set as just defined: there may very well be terms with a coefficient zero, or say accidental zeros; an instance presents itself, weight 10, where in the column $eg - bd^3$ we have $0\,ce^2$, no term in ce^2.

The changes actually required are very slight, viz.

Weight 11, instead of

$fg - b^3e^2$, we require $c^2h - b^3e^2$, old $fg - b^3e^2$, new named,
$c^2h - b^3d^2$, „ $fg - b^3d^2$, linear combination $(fg - b^3e^2) + 8(c^2h - b^3d^2)$,
$de^2 - b^3c^4$, „ $c^3f - b^3d^2$, old $de^2 - b^3d^2$, new named,
$c^3f - b^3c^4$, „ $de^2 - b^3c^4$, linear combination $(de^2 - b^3d^2) + 6(c^3f - b^3c^4)$.

Weight 12, instead of

$cf^2 - bcd^3$, we require $d^2g - bcd^3$, old $cf^2 - bcd^3$, new named,
$d^2g - c^3d^2$, „ $cf^2 - c^3d^2$, linear combination $(cf^2 - bcd^3) - 5(d^2g - c^3d^2)$;

but I have thought it desirable to give the complete tables for the weights in question, 11 and 12; and I have also rearranged the entire columns of the two tables so as to present in each of them the finals in AO. This is the case with the existing tables, except that there is a single transposition in the table weight 10.

Instead of the columns $cdf - b^4d^2$, $ce^2 - c^5$, we ought to have $ce^2 - c^5$, $cdf - b^4d^2$. The complete list up to the weight 12 is

$w = 2$: $c - b^2$
$w = 3$: $d - b^3$
$w = 4$: $e - c^2$, $e - b^4$
$w = 5$: $f - be^2c^2$, $cd - b^5$
$w = 6$: $g - d^2$, $ce - c^3$, $\cdots - b^6$
$w = 7$: $h - cf$, $\ldots - bc^3$, $\ldots - b^2c^2$, de, $c^2d - b^7$
$w = 8$: i, $df - e^2$, $cg - cd^2$, $\ldots - b^2d^2$, $c^2e - b^4c^2$, $\ldots - b^8$
$w = 9$: $j - bc^2$, $ch - d^3$, $dg - bcd^2$, $\ldots$, ef, $c^2f - bc^4$, $cde - b^2c^3$, $d^3 - b^3c^3$, $c^3d - b^9$
$w = 10$: bf^2, $ci - ce^2$, $dh - b^2c^2$, $eg - bd^3$, $f^2 - c^2d^2$, $ce^2 - c^5$, $cdf - b^4d^2$, $\ldots - b^{10}$
$w = 11$: bk, $di - b^4d^2e^2$, $eh - b^4$, $ef - b^5$, $c^2h - b^3e^2$, $\ldots$, $cdg - b^3cd^2$, $d^2f - bc^5$, $de^2 - b^3d^2$, $\cdots$, $c^3f - b^3c^4$, c^{11}
$w = 12$: $m - g^2$, $\ldots - cf^2$, $ei - c^3$, $\ldots - b^2f^2$, dj, $fh - bde^2$, $g^2 - c^2e^2$, $c^2i - b^2ce^2$, $ceg - d^4$, $d^2g - bcd^3$, $cf^2 - c^3d^2$, $cdh - b^4e^2$, $\ldots$, $def - b^3d^3$, $\ldots$, $c^2de - b^6e^2$, $d^4 - b^4d^2$, $\ldots$, $c^2df - b^6d^2$, $c^4e - b^8c^2$, $c^3d^2 - b^8c^2$, $c^6 - b^{12}$

The two new tables are as follows: some accidental numerical errors have been corrected.

TABLE, WEIGHT 11. (Complete table with 14 columns and 56 rows, double-page; cf. PDF pp. 242–243. The columns are headed (in CO/initial-term order): fg, bf^2, di, eh, c^2h, bcg, bdh, cdg, c^2f, cef, bf^2, b^3i, b^2dg, b^2ch, $\ldots$; the column totals are ±638, ±188, ±70, ±132, ±664, ±1640, ±57, ±170, ±43, ±278, ±192, ±35, ±98, ±48. The detailed entries reproduce those of the original printed tables.)

TABLE, WEIGHT 12. (Complete table with 21 columns and 77 rows, double-page; cf. PDF pp. 244–245. The column totals are ±1024, ±212, ±82, ±1740, ±2854, ±946, ±12, ±46, ±325, ±190, ±298, ±1664, ±442, ±18, ±52, ±51, ±258, ±156, ±48, ±68, ±32. The detailed entries reproduce those of the original printed tables.)

ON CLIFFORD'S PAPER "ON SYZYGETIC RELATIONS AMONG THE POWERS OF LINEAR QUANTICS."

[927]

From the *Proceedings of the London Mathematical Society*, vol. XXIII. (1892), pp. 99–104.

The paper in question, originally printed, *Proc. Lond. Math. Soc.*, t. III. (1869), pp. 9–12, is reproduced No. XIV., pp. 119–122, of the *Mathematical Papers* (8vo. Lond. 1882), where it is immediately followed by the paper No. XV., "On Syzygetic Relations connecting the Powers of Linear Quantics," pp. 123–129. The author, after referring to theorems in M. Paul Serret's *Geometrie de Direction* (Paris, 1869), proceeds as follows:—"By the use of Professor Sylvester's method of contravariant differentiation, I have arrived at certain extensions of these theorems, which I now proceed to explain," and he then states his Theorem I.: In order that a system of N points in a plane should all lie on a curve of the order n, it is sufficient that the pth powers of their distances from an arbitrary line should satisfy a linear homogeneous relation, the number N being given by the formula

$$N = \tfrac{1}{2}\alpha(n+3) + \tfrac{1}{2}(\beta+1)(\beta+2),$$

where α is the quotient, and β the remainder of the division of p by n, so that $p = \alpha n + \beta$, and $\beta < n$. And he then gives Theorem II., a like theorem as regards points in space; and, further, two Tables, A and B, for the values of N corresponding to given values of n and p in the two cases respectively.

Theorem I. is incorrect for the first value of N in Table A, viz. if $n = 1$, $p = 2$, then $N = 5$; the theorem here is: In order that a system of five points in a plane may lie in a line, the sufficient condition is that the squares of their distances from an arbitrary line shall satisfy a linear homogeneous relation; or, what is the same thing, if the squares of the distances of the five points satisfy a linear homogeneous relation, then the five points will lie in a line. The right conclusion is that four of the five points will lie in a line.

Before proceeding further, I slightly modify the form of the enunciation by defining (in the case of the plane figure) the power of a point to be its distance from an arbitrary line; or, what is the same thing, to be the nilfactum of the line-equation of the point; that is, for the point (x_1, y_1, z_1), the power is

$$\alpha x_1 + \beta y_1 + \gamma z_1,$$

where α, β, γ are arbitrary coefficients, or, if we please, line-coordinates. I say that $\alpha x_1 + \beta y_1 + \gamma z_1$ is the first power, $(\alpha x_1 + \beta y_1 + \gamma z_1)^2$ the second power, and so on. Clifford's Theorem I. thus is: If the pth powers of the N points satisfy a homogeneous linear relation, the N points are on a curve of the nth order.

In the case $n = 1$, $p = 2$, we have five points whose second powers satisfy a homogeneous linear relation; that is, if $(x_1, y_1, z_1), \ldots, (x_5, y_5, z_5)$ are the coordinates of the five points respectively, we have

$$\lambda_1(\alpha x_1 + \beta y_1 + \gamma z_1)^2 + \cdots + \lambda_5(\alpha x_5 + \beta y_5 + \gamma z_5)^2 = 0.$$

It is implicitly assumed that the points are distinct points, and we thus exclude such solutions as

$$\lambda_1 = \lambda_2 = \lambda_3 = 0, \quad (x_4, y_4, z_4) = k(x_5, y_5, z_5), \quad k^2\lambda_4 + \lambda_5 = 0.$$

Here α, β, γ are arbitrary, and the relation is equivalent to the six equations

$$\lambda_1 x_1^2 + \cdots + \lambda_5 x_5^2 = 0, \quad \lambda_1 y_1^2 + \cdots + \lambda_5 y_5^2 = 0, \quad \lambda_1 z_1^2 + \cdots + \lambda_5 z_5^2 = 0,$$

$$\lambda_1 y_1 z_1 + \cdots = 0, \quad \lambda_1 z_1 x_1 + \cdots = 0, \quad \lambda_1 x_1 y_1 + \cdots = 0;$$

and hence, eliminating the λ's, we have the relation

$$\begin{vmatrix} x_1^2, & y_1^2, & z_1^2, & y_1z_1, & z_1x_1, & x_1y_1 \\ x_2^2, & y_2^2, & z_2^2, & y_2z_2, & z_2x_2, & x_2y_2 \\ x_3^2, & y_3^2, & z_3^2, & y_3z_3, & z_3x_3, & x_3y_3 \\ x_4^2, & y_4^2, & z_4^2, & y_4z_4, & z_4x_4, & x_4y_4 \\ x_5^2, & y_5^2, & z_5^2, & y_5z_5, & z_5x_5, & x_5y_5 \end{vmatrix} = 0,$$

viz. this means that each of the determinants, obtained by selecting in any manner five out of the six columns, is $= 0$. This is a twofold relation, and thus it cannot be equivalent to the threefold relation

$$\begin{vmatrix} x_1, & y_1, & z_1 \\ x_2, & y_2, & z_2 \\ \vdots & \vdots & \vdots \\ x_5, & y_5, & z_5 \end{vmatrix} \text{[all minors } = 0]$$

which expresses that the five points are in a line.

But we see further that the twofold relation expresses that the five points are such that we can, through them and an arbitrary sixth point (x_6, y_6, z_6), draw a determinate conic; and this is the case only if four of the five points are in a line; viz. the conic is then the line-pair composed of this line and of the line joining the remaining two points. The right conclusion therefore is that, if the above linear relation is satisfied, then four of the five points lie in a line.

Clifford's proof is rather indicated than carried out, but, from the reference to contravariant differentiation, and from the second paper mentioned above, it must have been as follows:

Starting from the linear relation considered as an identity in (α, β, γ), and operating upon it with $p\partial_\alpha + q\partial_\beta + r\partial_\gamma$, p, q, r denoting arbitrary coefficients, we obtain

$$\lambda_1(px_1 + qy_1 + rz_1)(\alpha x_1 + \beta y_1 + \gamma z_1) + \cdots + \lambda_5(px_5 + qy_5 + rz_5)(\alpha x_5 + \beta y_5 + \gamma z_5) = 0;$$

hence, determining the ratios of p, q, r, say by the equations

$$px_4 + qy_4 + rz_4 = 0, \quad px_5 + qy_5 + rz_5 = 0,$$

and, instead of $\lambda_1(px_1 + qy_1 + rz_1)$, &c., writing A_1, &c., we have

$$A_1(\alpha x_1 + \beta y_1 + \gamma z_1) + A_2(\alpha x_2 + \beta y_2 + \gamma z_2) + A_3(\alpha x_3 + \beta y_3 + \gamma z_3) = 0,$$

a homogeneous linear relation between the first powers of the three points (x_1, y_1, z_1), (x_2, y_2, z_2), (x_3, y_3, z_3). We thus see that these points, say the points 1, 2, 3, are in a line; and, similarly, by different determinations of p, q, r, that the points 1, 2, 4 are in a line; and that the points 1, 2, 5 are in a line; that is, the points 3, 4 and 5 are each of them in the line 12 joining the points 1 and 2; that is, the points 1, 2, 3, 4, 5 are in a line.

But, if we examine the reasoning more closely, it appears that the first conclusion, 1, 2, 3 are in a line, depends on the assumption that neither 1, 2, nor 3 is in the line 45. Suppose, for instance, that 3 was in the line 45, the equations

$$px_4 + qy_4 + rz_4 = 0, \quad px_5 + qy_5 + rz_5 = 0,$$

imply $px_3 + qy_3 + rz_3 = 0$, and we have

$$A_1(\alpha x_1 + \beta y_1 + \gamma z_1) + A_2(\alpha x_2 + \beta y_2 + \gamma z_2) = 0,$$

satisfied by making the points 1 and 2 coincident. Thus, if 3 be on the line 45 (and similarly if 1 or if 2 be on the line 45), we cannot infer that 1, 2, 3 are in a line.

I assume, therefore, that neither 1, 2, nor 3 is in the line 45; we here conclude, as above, that 1, 2, 3 are in a line. Suppose for a moment that 5 is not on this line; 4 cannot be on each of the lines 15, 25, 35, and I assume, in the first instance,

that it is not on any one of these lines; thus the points 1, 2, 4 are no one of them on the line 35, and hence, by the like reasoning, 1, 2, 4 are in a line; that is, 4 is on the line 12, or, the points 1, 2, 3, 4 are in a line. If, however, 4 is on one of the lines 15, 25, 35, say it is on 35; then it is not on 25, and no one of the points 1, 3, 4 is on 25; hence, by the like reasoning, 1, 2, 4 are in a line. In this case, however, 4 being, by supposition, on the line 35, can only be the point 3; that is, 3 and 4 would be coincident, a case which need not be considered. The correct conclusion from the reasoning thus is, not that 1, 2, 3, 4, 5 are in a line, but that some four of these five points, say 1, 2, 3, 4, are in a line.

It is easy to see that, if we have on a line four points, then there exists between the second powers of these points a linear homogeneous relation. For let the distances of the points from a fixed point of the line be a, b, c, d respectively; and let the arbitrary line meet the line in a point O at a distance r from the fixed point. The distances of the four points from the point O are thus equal to $r+a$, $r+b$, $r+c$, $r+d$ respectively; and hence, writing k for the squared cosine of the inclination of the two lines, the second powers of the four points are $= k(r+a)^2$, $k(r+b)^2$, $k(r+c)^2$, $k(r+d)^2$ respectively; and we have thus, between the second powers p_1, p_2, p_3, p_4 of the four points, the homogeneous linear relation

$$\begin{vmatrix} p_1, & 1, & a, & a^2 \\ p_2, & 1, & b, & b^2 \\ p_3, & 1, & c, & c^2 \\ p_4, & 1, & d, & d^2 \end{vmatrix} = 0.$$

Moreover we can, in an infinite number of ways, form a linear combination

$$A_1 p_1 + A_2 p_2 + A_3 p_3 + A_4 p_4$$

of these second powers, which shall be independent of r, and have any given value whatever. We can therefore make this sum to be equal to the second power of any point 5 whatever (not on the line containing the four points) in relation to the arbitrary line; that is, given the four points in a line, and any other fifth point, we can establish between the second powers of these five points a linear homogeneous relation

$$A_1 p_1 + A_2 p_2 + A_3 p_3 + A_4 p_4 + A_5 p_5 = 0,$$

with non-evanescent values for each of the coefficients A. We thus see how, to the linear homogeneous relation between the second powers of the five points, there corresponds properly the non-symmetric relation: four of the five points are in a line.

The most simple cases are when $p = n$, viz. $p = n = 2$, $N = 6$; $p = n = 3$, $N = 10$, and generally $N = \frac{1}{2}(n+1)(n+2)$; and for these Clifford's theorem is easily verified. I have not examined the other cases, but it is probable that in each of them a correction is required.

ON THE ANALYTICAL THEORY OF THE CONGRUENCY.

[928]

From the *Proceedings of the London Mathematical Society*, vol. XXIII. (1892), pp. 185–188.

If the lines of a congruency are considered as issuing from the several points of a surface, or say as the quasi-normals of a surface, then the fundamental geometrical theory is established by an analysis closely similar to that for the theory of the curvature of a surface; viz. it is shown that each quasi-normal is intersected by two consecutive quasi-normals in two points respectively (corresponding to the centres of curvature), or say in two foci; we have on the surface two series of curves of quasi-curvature—only these do not in general intersect at right angles; the intersecting quasi-normals form two series of developable surfaces, each touching the surface of centres (or focal surface), along its cuspidal edge, &c.; and, in particular, each quasi-normal is a bitangent of the focal surface, touching it at the two foci respectively.

But the analysis assumes a very different form if we consider the congruency by itself, without thus connecting it with a surface. Regarding the congruency as determined by means of two equations,

$$U(a,b,c,f,g,h)=0, \quad V(a,b,c,f,g,h)=0,$$

between the six coordinates of a line ($af+bg+ch=0$, as usual), I take (a,b,c,f,g,h) for the coordinates of a particular line of the congruency, $(a_1,b_1,c_1,f_1,g_1,h_1)$ for those of a consecutive line. Denoting, for shortness, the derived functions

$$(\partial_a,\partial_b,\partial_c,\partial_f,\partial_g,\partial_h)U$$

by F, G, H, A, B, C, and similarly those of V by F', G', H', A', B', C', we have

$$a_1F+b_1G+c_1H+f_1A+g_1B+h_1C=0,$$

$$a_1F'+b_1G'+c_1H'+f_1A'+g_1B'+h_1C'=0;$$